PRODUCT STRATEGY

Algorithm Productization Strategy

A 7-Phase Roadmap from Research Code to Market-Ready Product

March 2026

Strategic Roadmap

Strategic Roadmap Overview

Seven phases from stabilization to market entry and long-term growth

01 Stabilize the Algorithm

Transform research script into reliable software component with clear interfaces and robust error handling

02 Build Cloud Execution Service

Secure API-based service protecting algorithm IP while enabling external lab access

03 Monetization Model

Two-tier pricing: Academic token-based plan and Enterprise custom license with priority support

04 Privacy Guarantees

No permanent storage, automatic deletion, encrypted communication—major selling point for labs

05 Enterprise Deployment

Optional on-premise licensed Docker container for pharma companies requiring full privacy

06 Market Entry Strategy

Start with Tier 2/3 universities for validation, then expand to pharma and biotech

07 Long-Term Vision

Platform expansion: batch screening, similarity search, workflow pipelines—becoming computational chemistry platform

Key Principles

Five core principles: IP protection, simplicity, market validation, reproducibility, organic expansion

01

PHASE 1

Stabilize the Algorithm

Timeline

1–2 weeks

Goal

Transform the research script into a reliable software component with production-grade quality

Key Actions

- **Understand code & dependencies** — Map all libraries, versions, and system requirements
- **Package into Python module** — Clean, installable package with proper structure
- **Define clear interface** — Standardized input/output with documentation
- **Add production features** — Logging, error handling, reproducibility, versioning

Input

Primary Input

SMILES string

Batch Processing

Batch file

Output

Structured Data

JSON format

Tabular Export

CSV format

Phase 1 Deliverable

A stable "analysis engine" that can run anywhere

No product yet—just a solid, reliable core ready for productization

PHASE 2

02 Build a Secure Cloud Execution Service

Goal

Make the algorithm usable by external labs without exposing the code —protecting IP while enabling access

Architecture Flow

Lab — External User
→ Request
API — FastAPI Gateway
→ Process
Container — Docker + Queue
→ Return
Lab — Results

Input Deleted — No storage

Key Principles

- **Algorithm stays on our servers** — IP never leaves secure infrastructure
- **Labs send input molecules** — Simple SMILES or batch file submission
- **Results returned immediately** — Real-time processing with minimal latency
- **No data stored after computation** — Privacy-first architecture

Benefits

- Protects algorithm IP
- Simple to deploy
- Easy to update algorithm

Technology Stack

Python	FastAPI
Docker	API Key

Phase 2 Deliverable

PHASE 3

03 Monetization Model

Strategy

Start simple. Payment integration can come later —first focus is usage validation and market fit.

Academic Plan

For universities & research labs

Token-Based Usage

Pay-per-use model with transparent pricing. Labs purchase token packages and consume based on actual usage.

Small Subscription

Optional monthly subscription for regular users with discounted rates and higher token allocations.

Affordable for Universities

Pricing designed for academic budgets—lower cost per analysis with educational discounts.

Ideal for: PhD research, course projects, small labs

Enterprise Plan

For pharma & biotech companies

Custom License

Negotiated licensing agreements tailored to organizational needs, compliance requirements, and usage patterns.

Higher Usage Limits

Scalable processing capacity for high-throughput screening, batch analysis, and large datasets.

Priority Support

Dedicated support channel, faster response times, technical consultation, and custom feature development.

Ideal for: Pharma R&D, biotech startups, CROs

Initial Focus	Payment Integration	Pricing Strategy
Usage Validation	Later Phase	Start Simple

PHASE 4

04 Privacy Guarantees

Privacy will be a major selling point for research labs

No Permanent Storage

Molecules are never stored permanently in our systems. All input data exists only in transient memory during processing.

- ✓ Data exists only in RAM during computation

Encrypted Communication

End-to-end encryption for all data transmission between labs and our servers using industry-standard protocols.

- ✓ TLS 1.3 encryption for all API communications

Automatic Deletion

Immediate and automatic deletion of all input data immediately after analysis completion.

- ✓ Zero retention policy with automated cleanup

Optional Local Deployment

For organizations with strict data sovereignty requirements, local deployment option available (Phase 5).

- ✓ On-premise option for maximum control

Why This Matters for Research Labs

Research labs work with proprietary molecules, pre-publication data, and competitive intelligence. Privacy guarantees remove adoption barriers and build trust—making it easier to convince labs to use our platform instead of building in-house solutions.

05

PHASE 5

Optional Enterprise Deployment

Target

For pharma companies that want full privacy and data sovereignty

What We Provide

A licensed Docker container that runs entirely within the client's infrastructure—no data ever leaves their systems.

Licensed Docker Container

Pre-configured, enterprise-grade container with all dependencies included. Deploys in minutes to any Docker-compatible infrastructure.

Important Constraint

We only build this after validating demand.

Enterprise deployment requires significant engineering effort for licensing, obfuscation, and support infrastructure. We build this only when:

- ✓ Multiple enterprise customers request on-premise option
- ✓ Revenue justifies development investment
- ✓ Cloud version is mature and stable

Key Features

Compiled/Obfuscated Core

Algorithm distributed as compiled binary or obfuscated code to protect IP while enabling local execution.

License Key Validation

Enterprise-grade license management with expiration, usage tracking, and renewal workflows.

No Data Leaving Infrastructure

Complete air-gapped operation. All processing happens locally with zero external communication.

Positioning

Enterprise Tier Premium Offering

06

PHASE 6

Market Entry Strategy

Phase 1: Initial Market

Start Here

Tier 2 / Tier 3 Universities

Regional universities, smaller research institutions, and specialized colleges with active chemistry/biology programs.

- **Lower Budget, High Curiosity**
  More willing to try new tools; less bureaucratic procurement
- **Easier Adoption**
  Faster decision cycles, less complex approval processes
- **Good for Testing & Feedback**
  Early adopters provide valuable product feedback

Goal: Validate product-market fit and gather testimonials

Phase 2: Expansion

After Validation

Pharma Research Groups

R&D departments at pharmaceutical companies, drug discovery teams, computational chemistry groups.

Biotech Startups

Agile biotech companies, AI-driven drug discovery startups, precision medicine firms.

Academic Labs with Grants

Well-funded research labs at Tier 1 universities, national laboratories, research institutes.

Goal: Scale revenue and establish market presence

Strategic Rationale

Starting with Tier 2/3 universities allows us to validate the product, refine the user experience, and build a track record before approaching enterprise customers. This phased approach reduces risk and ensures we have a mature product when entering competitive enterprise markets.

07

PHASE 7

Long-Term Vision

Vision Statement

If adoption grows, we expand the platform to become a lightweight computational chemistry platform for smaller labs

Platform Expansion

Batch Molecule Screening

High-throughput screening capabilities for large compound libraries with parallel processing and automated workflows.

Similarity Search

Find structurally similar molecules in databases using fingerprint comparisons and advanced similarity metrics.

Dataset Analytics

Comprehensive analytics dashboard for exploring molecular datasets, identifying patterns, and generating insights.

Workflow Pipelines

Customizable analysis pipelines that chain multiple algorithms and tools for complex computational workflows.

Ultimate Positioning

Computational Chemistry Platform

For Smaller Labs & Research Teams

- ✓ Accessible pricing vs. enterprise tools
- ✓ Easy to use, minimal setup required
- ✓ Cloud-first with privacy guarantees
- ✓ Scales from individual to team usage

Growth Strategy

Organic Expansion Based on User Needs

We don't build features in advance—we listen to real users and prioritize based on demand.

- → Feature requests drive roadmap
- → Validate before investing

Key Principles for the Project

Five core principles guiding the entire productization strategy

1 Protect Algorithm IP

Cloud execution first.

The algorithm never leaves our secure servers. External users interact only through APIs, ensuring complete IP protection while enabling broad access.

Core asset protection

2 Keep Infrastructure Simple

Minimal complexity, maximum reliability.

Use proven, battle-tested technologies. Avoid over-engineering. Simple systems are easier to maintain, debug, and scale.

FastAPI + Docker + Queue

3 Validate Market Before Scaling

Prove demand exists.

Don't build enterprise features until enterprise customers ask. Don't scale infrastructure until user growth justifies it.

Usage validation first

4 Build Reproducible Software

Scientific rigor matters.

Version control, dependency management, logging, and documentation. Results must be reproducible and auditable.

Research-grade quality

5 Expand Only After Real Users

Let user needs drive roadmap.

Don't build features in a vacuum. Listen to real users, understand their pain points, and prioritize based on actual demand.

User-driven development

Guiding Philosophy

These five principles ensure strategic discipline throughout the roadmap. They keep us focused on what matters: protecting our core asset, building reliable software, validating demand, and growing organically based on real user needs.

Executive Summary

The complete strategy in one sentence

> "My suggestion is we first **package the algorithm into a stable engine**, then **expose it through a secure cloud API for universities**, and **later offer an on-premise licensed version for pharma if demand appears**."

Phase 1–2

Build stable engine and secure cloud service—protect IP while enabling access

Phase 3–4

Launch with academic focus, validate market, establish privacy guarantees

Phase 5–7

Expand to enterprise if demand exists, grow platform capabilities organically

Architect-Level Thinking

This strategy combines technical rigor with business pragmatism. It protects core IP, validates demand before investing, and scales organically based on real market response.

Ready to Build

This roadmap provides a clear, phased path from research code to market-ready product—protecting IP, validating demand, and scaling strategically.

IP Protection

Cloud-first architecture keeps algorithm secure

Market Validation

Start simple, prove demand, then scale

Organic Growth

Expand based on real user needs

March 2026 • 7-Phase Strategy • Architect-Level Approach